

Get/Set Sensor Settings (Command 84h)

Command:

84h – NBF – ISB – [DB₁ - ... - DB_N] – CKS

Command Type:

Single Response Command

Response:

84h – NBF – ISB – DB₁ - ... - DB_N – CKS

Definitions:

84h - command byte

NBF - number of bytes to follow

ISB - Sensor Setting Byte Identifier (see table below)

DB₁ ... DB_N - Data bytes used to set and return the value of a particular sensor setting

CKS - checksum

Description:

This command is used to get and set the various sensor settings in the Capnostat module. When the command is transmitted to the Capnostat without the optional DB₁ ... DB_N data bytes, the current value of the specified sensor setting is transmitted in the command's response string. This corresponds to "getting" the current value of the sensor setting. If the DB₁ ... DB_N data bytes are transmitted from the host, the sensor setting is set to that value. This corresponds to "setting" the specified sensor setting, and this new value is transmitted in the command's response string.

A host must request a valid setting byte identifier – ISB. If an invalid ISB is requested, the command's response string will return with the ISB = 0 and no returned data bytes.

Examples:

- This example command string gets the current ETCO₂ Time Period:

84h – 02h – 05h – 75h

The Capnostat's response would be:

84h – 03h – 05h – 01h – 73h (current ETCO₂ Time Period is 1 breath)

- This example command string sets the current ETCO₂ Time to 10 seconds:

84h – 03h – 05h – 0Ah – 6Ah

The Capnostat's response would be:

84h – 03h – 05h – 0Ah – 6Ah (current ETCO₂ Time Period is 10 seconds)

The table below lists the ISB byte identifiers and the corresponding sensor settings.

ISB	Number of Bytes	Sensor Setting Description
0	0	<i>Invalid Instrument or Parameter Setting</i> Number of Data Bytes = 0
91	1	<i>Waveform Digital Filter</i> Default: On Conversion: DB ₁ = 0, Off DB ₁ = 1, On <i>This setting allows control of the 10Hz Digital Filter of the CO2 waveform.</i> <i>NOTE: This setting is automatically set to ON during Sleep Mode. Therefore, if Filtering is desired to be disabled (OFF), the command to disable Filtering must be resent after the command to exit Sleep Mode.</i> <i>NOTE 2: This setting is available in software version 2.0 and greater.</i>